India CPI Inflation Study - Answers

Source: *CPI_Inflation_Case_Study.xlsx* (Jan 2013 - May 2023, 124 months x 3 sectors x 27 categories). All figures below come from your workbook sheets (Insight1-5, Bonus Insights) and were re-verified against the cleaned *ConsumerPriceIndex* sheet.

Data preparation. Raw Data stores values as text with "NA" gaps. Rural Housing is missing for all months (not published separately) and April-May 2020 is missing for almost every category (lockdown: price collection suspended). The cleaned sheet has no blanks - gaps were imputed (moving-average / neighbouring-sector approach). The mapping of 27 original categories into 10 buckets (Food, Luxury, Clothing, Housing, Fuel and light, Health, Education, Transport and communication, Misc, General Index) is stored in *CPI-Unpivot*.

Q1. Contribution of broader categories to the CPI basket

Latest month = **May 2023**, sector Rural+Urban. Contribution = sum of index values of the categories in a bucket / total of all buckets (General Index excluded so shares add to 100%).

Bucket	Sum of index	Contribution
Food	2133.5	45.84%
Luxury	730.8	15.70%
Clothing	553.2	11.89%
Housing	350.8	7.54%
Health	185.7	3.99%
Fuel and light	182.8	3.93%
Education	177.1	3.81%
Misc	175.7	3.77%
Transport and communication	164.8	3.54%
Total	4654.4	100.00%

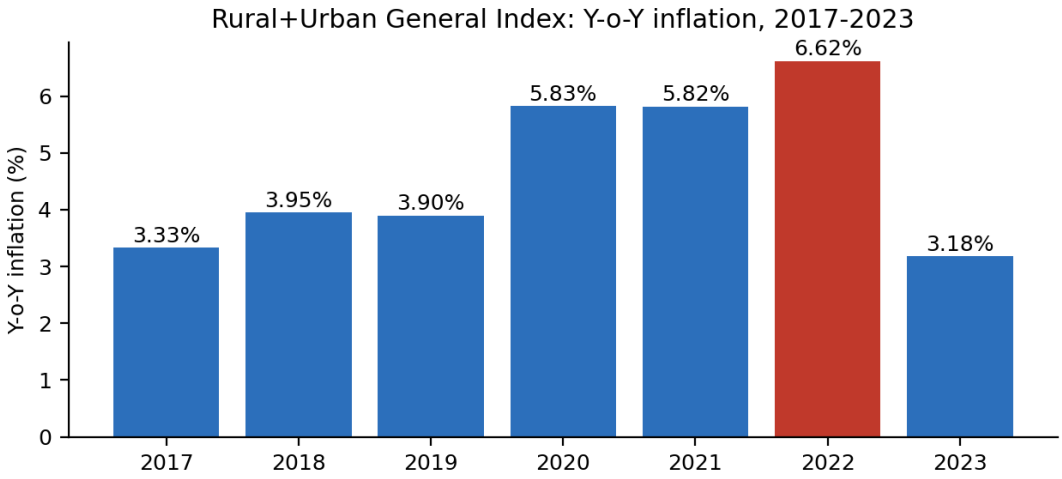
Answer: Food has the highest contribution (about 45.8%), nearly three times the next bucket (Luxury, 15.7%). Rural alone gives 45.2% and Urban 46.7%.

Note: your *Insight1* pivot shows Food at 44.4% - that is a *count* of rows (4,464 of 10,044) across all months, not an index-value share for the latest month. It gives the same conclusion, but the value-based table above matches the problem statement. Also, "Food and beverages" and "Clothing and footwear" are aggregates already in the mapping, so Food and Clothing are slightly inflated by double counting.

Q2. Y-o-Y inflation of the full basket (Rural+Urban General Index)

Y-o-Y = (average General Index of year - average of previous year) / previous year. Source: *Insight2*.

Year	Avg index	Y-o-Y %
2017	133.50	3.33
2018	138.78	3.95
2019	144.18	3.90
2020	152.58	5.83
2021	161.46	5.82
2022	172.15	6.62
2023*	177.62	3.18



Answer: 2022 had the highest inflation (6.62%) in the 2017-2023 window. (Rural alone peaked at 6.85% in 2022.) *2019 has 11 months and 2023 only Jan-May, so those annual averages are partial.

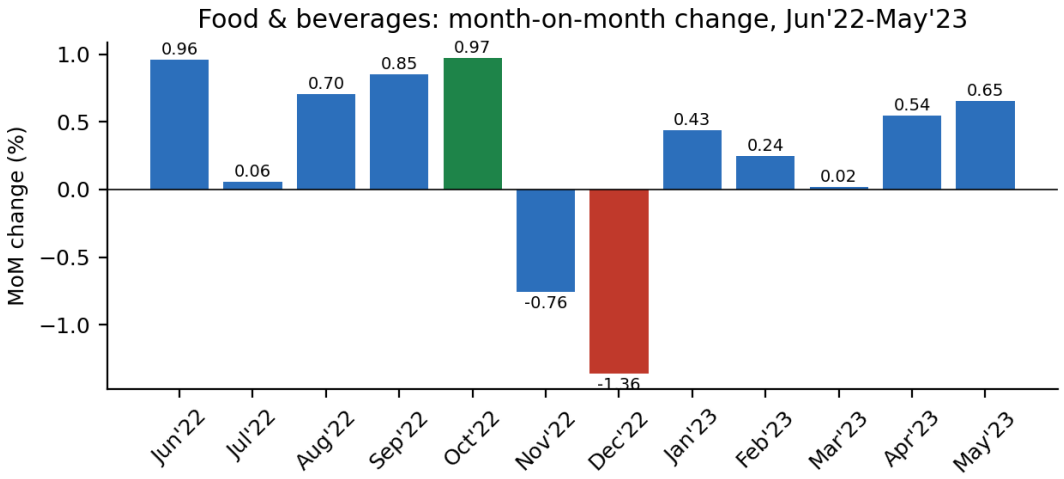
Why 2022? (research)

The 2022-23 Economic Survey says global inflation was fuelled by surging crude oil and commodity prices after the Russia-Ukraine war began, and that India's inflation peaked at 7.8% in April 2022. Ukraine is a major sunflower-oil exporter and India imports roughly 70% of its edible oil, so edible-oil inflation jumped (oils and fats +18.8% Y-o-Y in March 2022). Food and fuel prices rose together, a heat wave hit wheat output, and supply-chain disruption fed through to core items. The RBI began raising the repo rate from May 2022 and the government cut fuel duties, banned wheat exports and removed edible-oil import duties to contain prices.

Sources: Economic Survey 2022-23 (Ch. 5, indiabudget.gov.in); Acuite Ratings CPI note, Apr 2022; The News Minute / PTI, Apr 2022; Moneylife/IANS, Aug 2022.

Q3. Food prices, 12 months ending May 2023

Month-on-month change of the Food and beverages index (average of Rural, Urban and Rural+Urban, as in *Insight3*), June 2022 - May 2023.



Month	Jun'22	Jul'22	Aug'22	Sep'22	Oct'22	Nov'22	Dec'22	Jan'23	Feb'23	Mar'23	Apr'23	May'23
MoM %	0.96	0.06	0.70	0.85	0.97	-0.76	-1.36	0.43	0.24	0.02	0.54	0.65

Answer (a): Highest food inflation: October 2022 (+0.97% MoM; June 2022 was almost equal at +0.96%). Lowest: December 2022 (-1.36%). Prices climbed through autumn 2022 (vegetables +4.0% in Oct) then fell sharply in Nov-Dec as vegetable prices corrected (-8.6% and -12.7% in your Insight3 tracker) before rising again in spring 2023.

Absolute change in the index, May 2022 to May 2023, for each food category:

Food category	Change (pts)	Change %
Spices	+32.67	+17.7
Cereals and products	+19.30	+12.5
Milk and products	+14.53	+8.8
Prepared meals, snacks, sweets	+11.73	+6.4
Pulses and products	+11.20	+6.8
Egg	+11.00	+6.8
Sugar and Confectionery	+3.00	+2.5
Fruits	+1.17	+0.7
Meat and fish	-2.57	-1.2
Vegetables	-14.60	-8.1
Oils and fats	-31.37	-15.6

Answer (b): Spices is the biggest individual contributor to the rise (+32.67 index points, +17.7%), followed by Cereals (+19.30, +12.5%). Cereals is the largest rise among staples. Of the four items tracked in your Insight3 sheet, Milk rose the most (+14.5 pts). Vegetables (-14.6) and oils and fats (-31.4) pulled food inflation down. Net: Food and beverages rose from 173.87 to 179.67 (+3.3%).

Q4. Impact of COVID-19 (onset = March 2020)

Average month-on-month inflation for the 12 months before (Apr 2019 - Mar 2020) vs after (Apr 2020 - Mar 2021), plus annual Y-o-Y from *Insight4*. Index averaged across Rural, Urban and Rural+Urban.

Category	Avg MoM before %	Avg MoM after %	Change
General index	0.52	0.45	-0.07
Food and beverages	0.69	0.46	-0.23
Health	0.37	0.52	+0.15
Household goods and services	0.17	0.27	+0.10
Transport and communication	0.38	1.00	+0.62
Fuel and light	0.62	0.43	-0.19

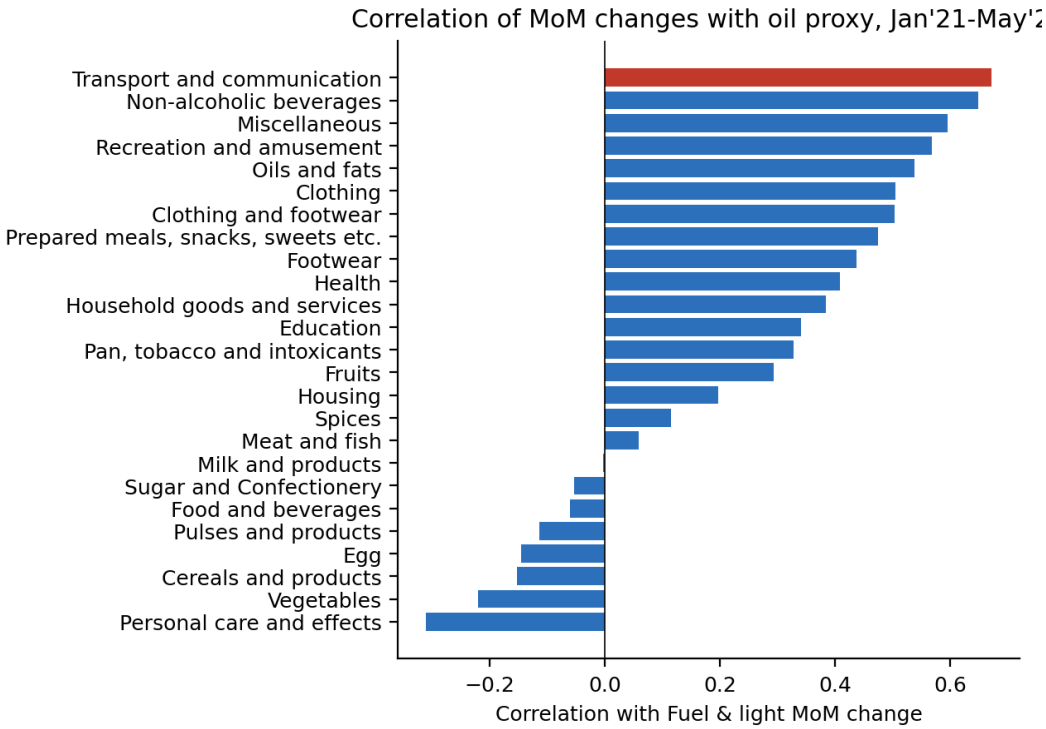
Annual Y-o-Y %	2018	2019	2020	2021
Food and beverages	1.57	4.24	7.77	4.58
Health	6.26	7.24	4.30	7.76
Household goods and services	5.06	4.18	2.19	5.23

Answer: Food was the visible shock: March 2020 saw a -1.06% MoM fall in food (vegetables -7.9%) as the first lockdown disrupted demand, but supply bottlenecks then pushed annual food inflation from 4.24% (2019) to **7.77% (2020)**, the highest of the three categories you tracked. **Health** inflation eased in 2020 (7.24% to 4.30%) but sped up afterwards (7.76% in 2021) and its monthly pace rose from 0.37% to 0.52% after March 2020. **Household goods and services** slowed in 2020 (2.19%) then rebounded in 2021 (5.23%). Transport and communication accelerated most after the lockdown (0.38% to 1.00% per month). Overall headline inflation rose from 3.90% (2019) to 5.83% (2020).

Caution: April-May 2020 prices were not collected, so those two months are imputed; the post-March results lean on interpolated points.

Q5. Imported oil price fluctuations and category inflation (2021-2023)

The dataset has no crude-oil price series, so **"Fuel and light" is used as the oil-price proxy** (as in your *Insight5*). Month-on-month % changes of every category (Jan 2021 - May 2023, 28 monthly changes) were correlated with the MoM change in Fuel and light using =CORREL. Your sheet computes the pairs Fuel-Transport (0.6718) and Fuel-Education (0.3415).



Rank	Category	Correlation
1	Transport and communication	0.672
2	Non-alcoholic beverages	0.649
3	Miscellaneous	0.596
4	Recreation and amusement	0.569
5	Oils and fats	0.539
...	Education (workbook comparison)	0.342
Lowest	Personal care and effects	-0.310

Answer: Transport and communication moves most strongly with oil/fuel price changes (r = 0.67). This is intuitive: fuel is a direct input to fares and freight. Education is only weakly linked (r = 0.34), showing it is largely insulated from fuel swings. Correlation does not prove causation, and Fuel and light is a domestic proxy that includes administered prices (LPG, electricity), not imported crude itself.

Level (not change) correlations are all above 0.9 because every index trends upward; use the MoM version above for a meaningful result.